

Effective Stiffness Tensor for the PML: A Transversely Isotropic Representation

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Motivations and Objectives

Motivations

The Perfectly Matched Layer (PML) can be viewed as a **coordinate transformation** of the original domain. The transformed medium possesses an effective stiffness tensor that governs wave absorption. Understanding this tensor is key to explaining **why PML can become unstable** in elastodynamics.

Objectives

Interpret the PML-induced stiffness tensor as a **transversely isotropic polar solid**.

Derive the strain energy, identify the block whose positive-definiteness is violated, and connect the result to **material-dependent stability conditions**.

Outline

- 1 Introduction
- 2 Form Invariance of the Elastic Wave Equation
- 3 Transversely Isotropic Polar Stiffness Tensor
- 4 Strain Energy Positivity and Stability
- 5 Analysis of Stability Condition
- 6 Conclusions

Navier's equation (frequency domain)

Find the displacement $\mathbf{u}$ such that

$$\operatorname{div} \mathbb{C}[\operatorname{grad} \mathbf{u}] + \rho \omega^2 \mathbf{u} = \mathbf{0} \quad (1)$$

where $\mathbb{C}$ is the fourth-order elastic tensor, ρ is the mass density, and ω is the angular frequency.

Key question: Under coordinate transformation $x^i \rightarrow X^I$, does this equation preserve its form?

- The answer depends on the **choice of gauge** — how the displacement vector is represented under the transformation.
- Two main approaches exist, each with distinct trade-offs.

Two Approaches to Transformation Elastodynamics

Let the Jacobian matrix be $\Lambda_I^i = \partial x^i / \partial X^I$.

Approach 1: Jacobian gauge (Λ_i^I)

$$\begin{aligned}\tilde{\mathbb{C}}_{IJKL} &= J^{-1} \Lambda_i^I \Lambda_j^J \Lambda_k^K \Lambda_l^L C^{ijkl} \\ \rho^{IJ} &= J^{-1} \Lambda_i^I \Lambda_j^J \delta^{ij} \rho\end{aligned}$$

- ✗ Transforms into **Willis form**
- ✓ Preserves symmetry of $\mathbb{C}$
- ✗ Density becomes **anisotropic tensor**

Approach 2: Identity gauge (δ_i^I)

$$\begin{aligned}\tilde{\mathbb{C}}^{IJKL} &= J^{-1} \Lambda_i^I \delta_j^J \Lambda_k^K \delta_l^L C^{ijkl} \\ \tilde{\rho} &= J^{-1} \rho\end{aligned}$$

- ✓ Preserves **Navier's form**
- ✗ $\mathbb{C}$ loses **minor symmetry**
- ✓ Density remains **scalar**

We adopt **Approach 2** (form invariance preserved) and interpret the loss of minor symmetry through the lens of **polar elasticity**.

Why a Polar Medium?

The identity gauge gives a tensor that preserves Navier's equation, but no longer acts like a classical Cauchy stiffness.

Classical Cauchy Solid

$$\psi = \psi(\varepsilon), \quad \varepsilon = \frac{1}{2} (\text{grad } \mathbf{u} + (\text{grad } \mathbf{u})^T).$$

- Energy depends only on symmetric strain
- Stress symmetry follows from minor symmetry

PML Effective Medium

$$\tilde{\mathbb{C}}^{IJKL} \neq \tilde{\mathbb{C}}^{JIKL}, \quad \tilde{\mathbb{C}}^{IJKL} \neq \tilde{\mathbb{C}}^{IJLK}.$$

- The tensor acts on the full gradient
- Skew stress may be generated

A polar interpretation keeps the transformed Navier form while giving a mechanical meaning to the non-Cauchy part of the PML stiffness.

Why Not a Full Cosserat Medium?

What we want to preserve

The PML transformation is interpreted within the **displacement-only Navier form**:

$$\operatorname{div} \tilde{\mathbb{C}}[\operatorname{grad} \mathbf{u}] + \tilde{\rho}\omega^2 \mathbf{u} = 0.$$

Thus, the only primary unknown is the displacement field $\mathbf{u}$.

What would change in a Cosserat interpretation

Introducing an independent microrotation ϑ changes the governing equation into a coupled system:

$$\sigma_{ji,j} = \rho \ddot{u}_i, \quad \chi_{ji,j} + \epsilon_{ijk} \sigma_{jk} = \iota \ddot{\vartheta}_i.$$

This no longer preserves the original displacement-only Navier form.

Interpretation adopted here

We therefore use a **polar effective-medium interpretation**: the PML stiffness acts on the full displacement gradient, and the skew stress is interpreted as an **effective body-torque density**, not as evidence of an independent microrotation degree of freedom.

Polar Energy for the PML Medium

Instead of forcing the PML tensor into a Cauchy form, we let the energy depend on the full displacement gradient:

$$\psi = \psi(e_{ij}), \quad e_{ij} = u_{i,j}. \quad (2)$$

The corresponding force stress is obtained by differentiating with respect to e_{ij} :

$$\sigma_{ij} = \frac{\partial \psi}{\partial e_{ij}} = a_{ijkl} e_{kl}. \quad (3)$$

Interpretation

The rotation vector introduced below is not an independent microrotation degree of freedom. It is only the axial-vector representation of the skew part of the displacement gradient.

Skew Stress as Induced Body Torque

The polar viewpoint gives a diagnostic meaning to the skew stress generated by the PML-transformed tensor.

$$\sigma_{ij} - \sigma_{ji} \neq 0 \implies c_i = \epsilon_{ijk} \sigma_{jk}. \quad (4)$$

Substituting the polar constitutive relation gives

$$c_i = \epsilon_{ijk} a_{jklm} e_{lm}. \quad (5)$$

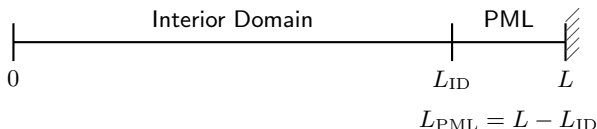
Consequence for PML

The body torque is not a new physical loading. It is a bookkeeping device that identifies how the minor-symmetry-breaking PML stiffness stores and transfers energy through skew stress.

PML as a Coordinate Transformation

When PML is applied in the x -direction only, the complex coordinate stretching yields a **diagonal** Jacobian:

$$\mathbf{\Lambda} = \text{diag}(a, 1, 1), \quad a = \alpha + i \frac{\beta}{\omega} \quad (6)$$



Since $\mathbf{\Lambda}$ is diagonal, the transformed stiffness $\tilde{\mathbb{C}}$ can be interpreted as a **transversely isotropic polar solid** with symmetry axis $\mathbf{e}_1$.

Displacement Gradient Decomposition

For a TI medium with symmetry axis $\mathbf{n}$, the displacement gradient $\mathbf{e} = \text{grad } \mathbf{u}$ decomposes into **six irreducible components**:

$$\mathbf{e} = \varepsilon_{nn} (\mathbf{n} \otimes \mathbf{n}) + \frac{1}{2} \varepsilon_{\perp} (\mathbf{I} - \mathbf{n} \otimes \mathbf{n}) + (\mathbf{q} \otimes \mathbf{n} + \mathbf{n} \otimes \mathbf{q}) + \mathbf{E}_{\perp} + \mathbf{W}. \quad (7)$$

Strain-like (sign change under reflection)

- $\varepsilon_{nn} = \mathbf{n} \cdot (\boldsymbol{\varepsilon} \mathbf{n})$
- $\varepsilon_{\perp} = \text{tr}(\mathbf{P} \boldsymbol{\varepsilon})$
- $\mathbf{E}_{\perp} = \mathbf{P} \boldsymbol{\varepsilon} \mathbf{P} - \frac{\varepsilon_{\perp}}{2} \mathbf{P}$
- $\mathbf{q} = \mathbf{P} \boldsymbol{\varepsilon} \mathbf{n}$

Rotation-like (invariant under reflection)

- $\omega_{\parallel} = \boldsymbol{\omega} \cdot \mathbf{n}$
- $\boldsymbol{\omega}_{\perp} = \mathbf{P} \boldsymbol{\omega}$

Mirror symmetry distinguishes these two classes — they **cannot couple** under centrosymmetry (with one exception).

Symmetry Restrictions via Schur's Lemma

Which components can couple in the strain energy? Classify by **SO(2) weight** m :

Weight m	Components	Rank	Coupling rule
0	$\varepsilon_{nn}, \varepsilon_{\perp}, \omega_{\parallel}$	scalar	Strain-strain: $\checkmark$, strain-rotation: $\times$
1	$\mathbf{q}, \omega_{\perp}$	vector	$\mathbf{q}$ - $\omega_{\perp}$ coupling: possible via $\mathbf{n} \times$
2	$\mathbf{E}_{\perp}$	tensor	Self-coupling only

Key insight

The cross product $\mathbf{n} \times \mathbf{q}$ converts $\mathbf{q}$ into an axial vector, reconciling its mirror symmetry with $\omega_{\perp}$. This enables the **shear-rotation coupling term** $\gamma \omega_{\perp} \cdot (\mathbf{n} \times \mathbf{q})$.

Collecting all allowed terms:

$$\psi = \underbrace{\frac{1}{2}A\varepsilon_{nn}^2 + B\varepsilon_{nn}\varepsilon_{\perp} + \frac{1}{2}C\varepsilon_{\perp}^2}_{\text{normal block}} + \underbrace{D\mathbf{E}_{\perp} : \mathbf{E}_{\perp}}_{\text{deviatoric}} + \underbrace{2L\mathbf{q} \cdot \mathbf{q} + \frac{1}{2}\alpha_{\perp}\boldsymbol{\omega}_{\perp} \cdot \boldsymbol{\omega}_{\perp} + \gamma\boldsymbol{\omega}_{\perp} \cdot (\mathbf{n} \times \mathbf{q})}_{\text{shear-rotation block}} + \underbrace{\frac{1}{2}\alpha_{\parallel}\omega_{\parallel}^2}_{\text{axial rot.}} \quad (8)$$

8 material parameters: $A, B, C, D, L, \alpha_{\parallel}, \alpha_{\perp}, \gamma$

- A, B, C, D, L : classical elastic moduli
- $\alpha_{\parallel}, \alpha_{\perp}$: rotational stiffness (polar elasticity)
- γ : **shear-rotation coupling** (unique to polar TI media)

Component Structure

The stiffness components are the Hessian of the quadratic energy:

$$a_{ijkl} = \frac{\partial^2 \psi}{\partial e_{ij} \partial e_{kl}}. \quad (9)$$

For the symmetry axis $\mathbf{n} = \mathbf{e}_1$, the stiffness components split into normal and shear families.

Normal Block

$$\begin{aligned} a_{1111} &= A, \\ a_{1122} &= a_{1133} = a_{2211} = a_{3311} = B, \\ a_{2222} &= a_{3333} = C + D, \\ a_{2233} &= a_{3322} = C - D. \end{aligned}$$

Shear Pair: (1, 2)

$$\begin{aligned} a_{1212} &= L + \frac{\alpha_{\perp}}{4} + \frac{\gamma}{2}, \\ a_{1221} &= a_{2112} = L - \frac{\alpha_{\perp}}{4}, \\ a_{2121} &= L + \frac{\alpha_{\perp}}{4} - \frac{\gamma}{2}. \end{aligned}$$

The (1, 3) shear pair has the same structure.

For a one-directional PML with $\Lambda = \text{diag}(a, 1, 1)$ and $J = a$, the transformed stiffness is

$$\tilde{C}_{IJKL} = \frac{1}{J} \lambda_I \lambda_K C_{IJKL}. \quad (10)$$

Normal Components

$$\begin{aligned} \tilde{C}_{1111} &= a(\lambda + 2\mu), \\ \tilde{C}_{1122} &= \tilde{C}_{2211} = \lambda, \\ \tilde{C}_{2222} &= \tilde{C}_{3333} = \frac{\lambda + 2\mu}{a}, \\ \tilde{C}_{2233} &= \tilde{C}_{3322} = \frac{\lambda}{a}. \end{aligned}$$

Shear Components

$$\begin{aligned} \tilde{C}_{1212} &= a\mu, \\ \tilde{C}_{1221} &= \tilde{C}_{2112} = \mu, \\ \tilde{C}_{2121} &= \frac{\mu}{a}, \\ \tilde{C}_{2323} &= \tilde{C}_{2332} = \frac{\mu}{a}. \end{aligned}$$

Matching PML to TI Polar Parameters

Parameter	Expression	Physical meaning
A	$a(\lambda + 2\mu)$	Axial P-wave modulus
B	λ	Normal coupling (unchanged)
C	$(\lambda + \mu)/a$	In-plane bulk modulus
D	μ/a	In-plane shear modulus
L	$\mu \frac{(a+1)^2}{4a}$	Average shear stiffness
$\alpha_{\parallel}$	0	No axial rotational stiffness
$\alpha_{\perp}$	$\mu \frac{(a-1)^2}{a}$	Transverse rotational stiffness
γ	$\mu \left(a - \frac{1}{a} \right)$	Shear-rotation coupling

When $a = 1$ (no PML): $\alpha_{\perp} = 0$, $\gamma = 0 \Rightarrow$ recovers standard isotropic elasticity.

When $a \in \mathbb{C}$: polar terms activate $\Rightarrow$ minor symmetry breaks.

Block Decomposition of Strain Energy

The energy ψ decomposes into **four independent blocks**:

(i) Normal block

$$\frac{1}{2} \begin{bmatrix} \varepsilon_{nn} & \varepsilon_{\perp} \end{bmatrix} \begin{bmatrix} A & B \\ B & C \end{bmatrix} \begin{bmatrix} \varepsilon_{nn} \\ \varepsilon_{\perp} \end{bmatrix}$$

(ii) Transverse deviatoric

$$D \mathbf{E}_{\perp} : \mathbf{E}_{\perp}$$

(iii) Axial rotation

$$\frac{1}{2} \alpha_{\parallel} \omega_{\parallel}^2 = 0 \quad (\alpha_{\parallel} = 0)$$

(iv) Shear-rotation block

$$\frac{1}{2} \begin{bmatrix} \varepsilon & \omega \end{bmatrix} \begin{bmatrix} 4L & \pm\gamma \\ \pm\gamma & \alpha_{\perp} \end{bmatrix} \begin{bmatrix} \varepsilon \\ \omega \end{bmatrix}$$

Stability requires each block to be **Hermitian positive semi-definite** (since $a \in \mathbb{C}$).

Blocks (i)–(iii) can be made positive. Block (iv) is the **critical** one.

The Shear–Rotation Determinant is Always Non-Positive

The Hermitian part of the shear–rotation block gives:

$$\mathbf{H}_S = \begin{bmatrix} 4\Re L & \Re \gamma \\ \Re \gamma & \Re \alpha_{\perp} \end{bmatrix} \quad (11)$$

After substituting $a = \alpha + i\beta/\omega$ and expanding:

$$\det(\mathbf{H}_S) = (4\Re L)(\Re \alpha_{\perp}) - (\Re \gamma)^2 = -\frac{4\mu^2 \beta^2}{\omega^2 \alpha^2 + \beta^2} \leq 0 \quad (12)$$

Conclusion

Whenever $\beta \neq 0$ (PML attenuation active), the shear–rotation block **cannot** be positive semi-definite.

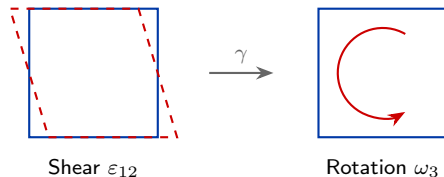
⇒ The PML effective medium admits **local energy divergence** in the shear–rotation sector.

Connection to Energy Gain Instability (Loh et al., 2009)

Loh et al. showed that PML applied to backward-wave structures acts as an **anisotropic medium with a gain direction**, producing net amplification instead of attenuation.

Our result provides a **local elastodynamic analogue**:

- The indefinite shear-rotation block is the **gain channel** through which modal energy amplification enters the PML.
- This is intrinsic to the polar structure — not a numerical artifact.



$\det(\mathbf{H}_s) = 0$ only when
 $a = 1$ (no PML) or $a \in \mathbb{R}$ (no attenuation).

Previous Empirical Stability Condition (Goh et al., 2025)

From eigenvalue analysis, the following empirical condition was proposed:

$$\left| \operatorname{Re}\{\hat{L}\} \right| > C \left| \operatorname{Im}\{\hat{L}\} \right|, \quad 0 < C \leq 1 \quad (13)$$

Physical interpretation: the wave must propagate sufficiently within the PML before being attenuated too rapidly.

For $C = 1$, this gives:

$$\begin{cases} \alpha \geq \frac{\beta}{\omega} - 1, & \text{for classical PML,} \\ \alpha \geq \frac{\beta(\omega - \omega_0)}{\omega^2 + \omega_0^2} - 1, & \text{for CFS-PML.} \end{cases} \quad (14)$$

Question: Can we derive this from the effective medium framework?

Deriving Stability from the Normal Block

The normal block positive semi-definiteness requires:

$$\mu(3\lambda + 2\mu) \alpha^2 \geq \lambda^2 \frac{\beta^2}{\omega^2} \quad (15)$$

For the classical PML ($\omega_0 = 0$), specializing to the PML profile $\lambda(s) = 1 + \alpha\xi^m$ with $\xi = (s - L_{ID})/L_{PML}$:

$$1 + \alpha\xi^m \geq \kappa \frac{\beta\xi^m}{\omega} \quad (16)$$

$$\text{where } \kappa = \frac{\lambda}{\sqrt{\mu(3\lambda + 2\mu)}} = \frac{2\nu}{\sqrt{2(1 + \nu)(1 - 2\nu)}}$$

The most restrictive condition occurs at $\xi = 1$ (PML outer edge):

$$\alpha \geq \kappa \frac{\beta}{\omega} - 1 \quad (17)$$

Extension to CFS-PML

For CFS-PML ($\omega_0 \neq 0$), the same procedure yields:

$$\alpha \geq \frac{\beta(\kappa\omega - \omega_0)}{\omega^2 + \omega_0^2} - 1 \quad (18)$$

Comparison with empirical condition

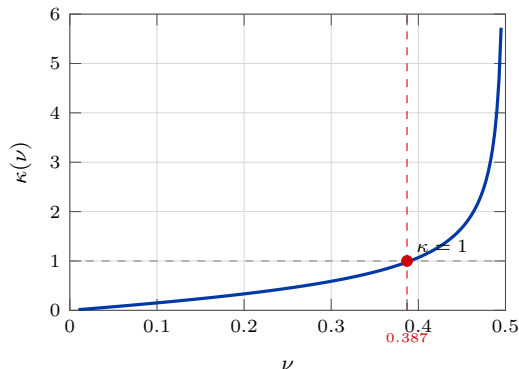
	Empirical (Goh et al.)	This work
Classical	$\alpha \geq \frac{\beta}{\omega} - 1$	$\alpha \geq \kappa \frac{\beta}{\omega} - 1$
CFS	$\alpha \geq \frac{\beta(\omega - \omega_0)}{\omega^2 + \omega_0^2} - 1$	$\alpha \geq \frac{\beta(\kappa\omega - \omega_0)}{\omega^2 + \omega_0^2} - 1$

The **only difference** is the material parameter κ . The empirical condition corresponds to $\kappa = 1$.

The Material Parameter $\kappa(\nu)$

Expressing κ in terms of Poisson's ratio:

$$\kappa(\nu) = \frac{2\nu}{\sqrt{2(1+\nu)(1-2\nu)}} \quad (19)$$



- $\kappa = 1$ at $\nu \approx 0.387$.
- $\nu < 0.387$: $\kappa < 1$
 $\Rightarrow$ empirical condition is **more restrictive** than necessary.
- $\nu > 0.387$: $\kappa > 1$
 $\Rightarrow$ empirical condition may be **insufficiently restrictive**.

Most metals/concrete: $\nu < 0.387$

$\Rightarrow$ empirical condition includes a safety margin.

Rubber, near-incompressible: $\nu \rightarrow 0.5$

$\Rightarrow$ risk of unexpected instability.

Practical Implications: Steel vs. Rubber

Steel ($\nu = 0.30$, $\kappa = 0.588$)

$\kappa < 1$: the material-dependent condition is **less restrictive** than the empirical one.

$\Rightarrow$ The empirical criterion $|\operatorname{Re}\{\hat{L}\}| > |\operatorname{Im}\{\hat{L}\}|$ provides an **extra safety margin**.

May predict instability even when the PML is actually stable.

Rubber ($\nu = 0.49$, $\kappa = 4.014$)

$\kappa > 1$: the material-dependent condition is **more restrictive** than the empirical one.

$\Rightarrow$ The empirical criterion may **incorrectly predict stability**.

Numerical divergence can occur even when $|\operatorname{Re}\{\hat{L}\}| > |\operatorname{Im}\{\hat{L}\}|$ is satisfied.

See Figure 1 in the paper for the comparison plots ($\hat{A} = 0$, $\hat{B} = 10$; steel with $\hat{\omega}_o = 1.5$ vs. rubber with $\hat{\omega}_o = 5$).

- The PML can be interpreted as a **transversely isotropic polar medium**.
- The strain energy splits into four independent blocks. The **shear-rotation block** always violates positive semi-definiteness when $\beta > 0$:

$$\det(\mathbf{H}_S) = -\frac{4\mu^2\beta^2}{\omega^2\alpha^2 + \beta^2} \leq 0$$

- The **normal block** yields a material-dependent stability condition involving $\kappa(\nu) = 2\nu/\sqrt{2(1+\nu)(1-2\nu)}$:

$$\alpha \geq \kappa \frac{\beta}{\omega} - 1 \quad (\text{classical PML})$$

- This closely matches the empirical condition of Goh et al., with the **only difference being** κ .
- Additional simulations are required for validation.
Further analysis of the remaining blocks should be conducted.



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